# Niels Uitterdijk

Experienced & entrepreneurial cloud architect driving innovation with a flair for AKS & machine learning platforms.

## Profile

Certified architect with a five years of team leading experience in cloud & data infrastructure design across heavy industry, FSI, and the telcom sector. I'm a driven engineer, entrepreneur and deeply committed to innovation and pragmatism. My expertise in cloud engineering is backed by a diverse project portfolio spanning IIoT, machine learning, and generative AI. I excel in designing scalable, secure, and efficient cloud solutions to meet business needs and enhance operational efficiency.

## Most Proud Of

- Successfully founding a startup
- Standardizing AKS migration for dozens applications for UBS
- Building an award winning electric racecar

## Strengths

Problem-Solving · Cloud Architecture · ML Platforms · Security · Leadership · Systems Design · Project Mgtm. · Customer Focus · Analytical · Resilience · DevOps

## Languages

| | |
|---|---|
| Dutch | Native |
| English | Fluent |
| German | B1 |

## Certifications

(CKA) Certified Kubernetes Administrator
AZ-104 Azure Administrator
AZ-305 Solutions Architect Expert

## Contact


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## Professional Experience

### Expert Cloud Engineer

Jun 2023 – ongoing | Zühlke Engineering AG | Zurich

- Led the **Cloud Architecture** for multiple large-scale solutions, from GPT-4 landing zones, strategic AKS migration platforms and full IIoT solutions.
- Succesfully provided strategic **Cloud Consulting** for governance, migration & vulnerability management.
- **Bids & Sales:** Proactively pushed sales activities in the IoT sector, leading the technical side of various bids, including the provisional architecture for a 1.5M CHF RfP.
- **Technical Execution:** Actively developed IaC with Terraform, Ansible, Bicep, (a.o.) to configure, automate and secure infrastructure provisioning & drive DevOps practices.
- **Team / Deliver Lead:** took ownership of the delivery and guided or led the teams through various projects.

### Founder & CTO

Mar 2020 – Mar 2023 | Amplo AG | Zurich

- Founded a tech startup, secured 1.6M CHF funding, hired seven engineers and launching multiple products.
- Developed an event-driven predictive maintenance platform, automating 90% of the RCA tickets for various clients.
- Established a strong culture of collaboration, innovation, and customer-focus, without having any employee quit during the tougher times.
- Served over 1'000 predictive models through an inhouse developed AutoML pipeline, supporting the various products.

### Cloud & ML Freelancer

Oct 2019 – Aug 2020 | Upwork | Remote

Acquired 'Rising Talent' status, with a rating of 4.92 / 5.0

### Machine Learning Intern

Feb 2019 – Aug 2019 | Tritium Ltd. Pty. | Brisbane

Analysed failure data and deployed predictive models in AWS lambda, connected with telemetry in S3 to improve successful charger rate by ±5 by catching two failures prematurely.

### Powertrain Department Manager

Jul 2016 – Aug 2017 | Formula Student Team Delft | Delft

Leading my department of eleven engineers to design & build the battery, motorcontrollers & motors of our 4WD electric racecar, Competed in FS Hungary (2nd), FS Spain (4th) and FS Germany (12th).

### Teaching Assistant Advanced Dynamics

Aug 2015 – Feb 2016 | Technical University of Delft | Delft

## Education

### MSc Systems & Control

Sep 2017 – Feb 2020 | Technical University of Delft

MSc thesis on online distributionally robust optimization for fault detection for Tritium's 5'000 public DC Fast chargers (Grade 8.5 / 10.0).

### BSc Mechanical Engineering

Sep 2013 – Jul 2016 | Technical University of Delft

## Extra-curricular

### Board - Secretary

Sep 2016 – Aug 2017 | SRC Thor | Delft

### Exchange Minor

Aug 2015 – Feb 2016 | Lund University | Lund, Sweden

### BSc Honours Programme

Sep 2014 - Aug 2015 | Technical University of Delft

Elected from the top 5% students.

## Skills

### Cloud Platforms
Azure (4y), GCP (1y), AWS (3m)

### Security & Governance
Sentinel, Policy, Zero Trust

### Architecture & Design
High Availability, Disaster Recovery, Autoscaling, Networking

### Infrastructure as Code
Terraform, Bicep, ARM, Ansible

### DevOps & Automation
CI/CD, Gitlab, Github, Azure DevOps, ArgoCD, Ansible

### Data & AI
SQL, Data Lakes, MLOps, Messaging & Event streaming, GPT

## Notable Projects

### Develop Strategic AKS Infra & Automation
May '24 - ongoing | 6 months | Senior Cloud Engineer & Tech Lead

AKS | Helm | Ansible | Terraform | Istio | Gitlab

A leading international Swiss bank is merging two application landscapes whilst migrating from PCF to Azure. Our Cloud Adoption team built the bridge between the operations, strategy, security & application teams for the dedicated AKS migration path. We aligned application requirements and monitoring concept while centralizing the infrastructure configuration & provisioning layer. We started with the Ansible building blocks provided by the platform team but later rebuild with Terraform. Our Principal Architect got sidetracked on another initiative, at which point I took over the tech lead role over the five engineers working on this project.

### Azure Cloud Security & Governance Concept
Oct '23 - Apr '24 | 7 months | Cloud Lead

Sentinel | Terraform | Policy | Governance | Security | Cloud Migration

For an upcoming FINMA audit, I led a two person engagement to advise on the Cloud Governance & Security Monitoring concepts for a leading Swiss health insurer. We oversaw the resolution of 1,600 vulnerabilities in two months and supported the creation of their cloud governance, cloud security and cloud operations team. We initiated Sentinel and Policy setups, boosting their Secure Score by 12% and enhancing centralized monitoring for steady security.

### GPT-4 Landing Zone & Underwriter Assistant
Oct '23 - Aug '24 | 10 months | Cloud Lead

OpenAI | Azure DevOps | AKS | Terraform | SQL Database | .NET

For a global insurance firm, we started prototyping a GPT-4 powered application to streamline the underwriting process. After carefully deploying this to production and integrating it into their proprietary underwriting platform, we turned our IaC into a full-fletched landing zone for GPT powered applications.

### IIoT Cloud Architecture for Medical Machines
Nov '23 - Jan '24 | 3 months | Cloud Architect

Azure B2C | HA/DR | Front Door | AKS | IoT Hub | Stream Analytics | Cloud Migration

To boost revenue and outpace competitors, a Swiss medical manufacturer aimed to collect machine data. We architected their Landing Zone, IIoT infrastructure, and Azure B2C identity management, addressing regulatory compliance, secure networking, and scalable architecture. This preparation enabled a smooth implementation start.

### IoT Platform for EV Chargers
Jun '22 - Apr '23 | 10 months | Tech Lead

AKS | IoT Hub | Data Lake | Functions | Postgres | RabbitMQ | Python, TS | Azure DevOps

We developed an charger management platform to tackle EV charger reliability. It centralizes documentation, configurations, live telemetry, and uses machine learning for failure detection, accelerating root cause analysis by up to 80%.

### Smart Maintenance Platform for Industrial Machinery Manufacturers
Jan '21 - Jun '22 | 1.5 years | Tech Lead

GKE | GCP Dataflow | GCP IoT Core | Azure | Python, JS | Github Actions

Designed and implemented a machine learning platform for predictive maintenance, integrating IoT data sources. Enabled service engineers to efficiently manage data and develop over 1,000 machine learning models for on-premises and cloud deployment, successfully predicting dozens of daily failures.

### Solar Manager AG Underperformance Detection
2020 | 6 months | Machine Learning Engineer

AWS EC2 | AWS S3 | Python | Github Actions

Solar panel degradion often happens unnoticed, causing energy losses. We developed a model to check whether the daily energy prodution was as expected, given the weather conditions. We detected underperformance in 2% of users and increased user engagement by 15%.

### Saudi Aramco Processhub
2019 | 9 months | Machine Learning Engineer

LNG transportation requires a large industrial compressor for liquification which is hard-to-control and a single point of failure. We build a recommendation engine using a predictive model to support the control engineers, reducing flaring by 9%.